

Projects 2 (OpenSSL):

In this project you will learn how to use OpenSSL in the command line for basic functions.

- I. Please watch YouTube video, the link is given below and write detail report what you have learned from the tutorial
<https://www.youtube.com/watch?v=-nEh7X4dtuw>
- II. Do the following easy tasks and write the detail report how you get your results. ***In your report, please include snapshot of output with you source code and detail explanation.***
- III.

Tasks

1. Getting started: Start your PC (Windows or Linux) with an OpenSSL installation
 - a. Start the OpenSSL command line
\$ openssl
 - b. List commands by type
> list -commands
> list -cipher-commands
> list -digest-commands
 - c. Use the help to find out more about OpenSSL
> help
2. Performance of OpenSSL
 - a. Make a speed test on your PC-platform with the speed command
> speed
 - b. Compare the results for symmetric encryption (e.g., AES-CBC) and RSA signature.
Example:
>openssl speed rsa2048
3. Using OpenSSL from the command line interface
 - a. Create a text file with some input and encrypt it using
 - i. AES-128 CBC
 - ii. AES-256 CTR
 - iii. DES
 - b. Create a 2048 bit RSA public and private key
4. Exchange of encrypted data.
 - a. Encrypt a file (e.g., a text file) with an algorithm and a key length of your choice.
 - b. Exchange the file and the necessary credentials for decryption (i.e., algorithm, key) with your neighbor.
 - c. Decrypt the secret of your neighbor.

